VIBECHECK: DISCOVER & QUANTIFY QUALITATIVE DIFFERENCES IN LARGE LANGUAGE MODELS

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ABSTRACT

Large language models (LLMs) often exhibit subtle yet distinctive characteristics in their outputs that users intuitively recognize, but struggle to quantify. These "vibes" – such as tone, formatting, or writing style – influence user preferences, yet traditional evaluations focus primarily on the singular vibe of correctness. We introduce **VibeCheck**, a system for automatically comparing a pair of LLMs by discovering identifying traits of a model ("vibes") that are well-defined, differentiating, and user-aligned. VibeCheck iteratively discovers vibes from model outputs and then utilizes a panel of LLM judges to quantitatively measure the utility of each vibe. We validate that the vibes generated by VibeCheck align with those found in human discovery and run VibeCheck on pairwise preference data from real-world user conversations with Llama-3-70b vs GPT-4. VibeCheck reveals that Llama has a friendly, funny, and somewhat controversial vibe. These vibes predict model identity with 80% accuracy and human preference with 61% accuracy. Lastly, we run VibeCheck on a variety of models and tasks including summarization, math, and captioning to provide insight into differences in model behavior. VibeCheck discovers vibes like Command X prefers to add concrete intros and conclusions when summarizing in comparison to TNGL, Llama-405b often overexplains its thought process on math problems compared to GPT-4o, and GPT-4 prefers to focus on the mood and emotions of the scene when captioning compared to Gemini-1.5-Flash. Code and vibe visualizer found at https://bench-mark.org/

1 INTRO

> **vibe check :** A process by which a group obtains a subjective assessment of another person, place, or thing. – *Urban Dictionary*

How a large language model writes a story, explains a concept, or edits an essay can be evaluated along many different dimensions such as creativity, formatting, and writing style. However, most evaluations focus on one dimension: *"correctness"*. State-of-the-art in evaluation methods remain largely focused on measuring accuracy for question answering and analytical reasoning tasks (Hendrycks et al., 2021a; Wang et al., 2019b;a; Hendrycks et al., 2021c), and methods which aim to provide a more holistic view of LLMs (Zhang et al., 2024; Padlewski et al., 2024; Mehri & Eskenazi, 2020b) rely on predefined concepts like conciseness, clarity, and trustworthiness to measure a model's performance. These evaluation approaches fail to capture the open-ended nature of LLM applications and the critical dependence on subjective user preferences and context of the task. For instance, tone and creativity might be crucial in creative writing, whereas efficiency and readability are crucial in coding tasks. To best inform users of which model would be best for their needs, we require flexible evaluation methods that can both *discover* and *measure* the relevant axes to evaluate for a given task.

When interacting with a set of LLMs for an extended period, a user can often tell which model generated a particular response by looking at certain traits of the outputs. We define these identifying traits of models as *"vibes"*. For instance, users have found Llama-3 outputs tend to be more friendly compared to outputs from GPT-4 and Claude which tend to be more formal (see Figure 1); in other words, Llama-3 ranks high on the friendliness vibe, defined by the axis `formal` $\rightarrow$ `friendly`. Using these insights, we might select Llama for customer service tasks and Claude for coding tasks.

Understanding these vibes helps inform the development and deployment of models, but discovering and validating them for each model can be time-consuming and difficult. To address this, we outline how one can find and, more importantly, measure an LLM's vibe by formalizing three necessary and quantifiable traits of a useful vibe: *well-defined* (agreement among multiple users), *differentiating* (ability to distinguish between models), and *user-aligned* (predictive of user preferences).

We introduce **VibeCheck**, a system which qualitatively analyzes pairs of models by automatically finding well-defined, differentiating, and user-aligned vibes. Motivated by recent work in using LLM's in lieu of human judgment (Zheng et al., 2023; Zhang et al., 2024; Zhong et al., 2023; 2022; Dubois et al., 2023), VibeCheck models the qualitative analysis process by identifying the axes on which these model outputs differ to obtain a core set of vibes (e.g friendliness). Once these vibes are obtained, VibeCheck employs a panel of LLM judges (Verga et al., 2024) to determine where each model's output falls on this vibe (e.g. more formal or more friendly) in order to obtain numeric scores which are then used to measure a vibe on each of our 3 key criteria.

We run VibeCheck on several datasets to evaluate its effectiveness across different scenarios in Section 5. First, we validate that the vibes discovered by VibeCheck align well with human-annotated differences between ChatGPT and human responses using the Human ChatGPT Comparison Corpus (HC3). Next, we demonstrate that VibeCheck outperforms a predefined list of vibes in predicting user preferences on real-world comparison data from Chatbot Arena, achieving 80% accuracy at predicting model identity and 61% accuracy and predicting user preference. Inspecting the vibes of VibeCheck, we find that Llama-70b uses more typographic emphasis, more examples, and is funnier than GPT-4 and Claude-3-Opus. Conversely, we find that GPT-4 and Claude comment much more on ethics and limitations than Llama, which is more willing to give controversial responses.

Lastly, in Section 6 we apply VibeCheck to several applications: text summarization on CNN/Daily-Mail, math problem-solving on MATH, and image captioning on COCO. Using VibeCheck, we find insightful qualitative differences between models with similar accuracy on correctness metrics but differing user preferences. For instance, Command X prefers to add concrete intros and conclusions when summarizing in comparison to TNGL, Llama-405b often overexplains its thought process on math problems, and GPT-4 prefers to focus on the mood and emotions of the scene when captioning.

2 Related Work

Aspect-based evaluations. The number of benchmarks in the NLP community has exploded in recent years, with a growing body of work on exploring a more holistic evaluation of language models. Several works (Pang et al., 2020; Banerjee & Lavie, 2005; Sellam et al., 2020) aim to improve on automatic metrics like BLEU (Papineni et al., 2002) and ROUGE (Lin, 2004) scores to better measure how well a models output aligns with the ground truth by incorporating more nuanced evaluation criteria like factual accuracy, fluency, and conciseness. Similarly, efforts have been made (Liang et al., 2023; bench authors, 2023; Kiela et al., 2021; Wang et al., 2019b;a) to standardize model evaluation by evaluating models on many of these metrics across various tasks.

Moving away from measuring model outputs on ground truth responses, work from Mehri & Eskenazi (2020b); Zhang et al. (2024); Li et al. (2019); Mehri & Eskenazi (2020a); Gehrmann et al. (2021) evaluate model outputs on criteria like helpfulness and clarity using LLM judges on more open ended tasks like dialogue, role-play, and summarization. While these efforts supply a great foundation for measuring correctness, they all define the axes on what makes something correct beforehand. In contrast, VibeCheck aims to automatically discover these axes (vibes) and verify their utility to the user by measuring the correlation between vibes and human preference.

Pairwise comparison of LLMs. HCI tools like Google's AutoSxS (Google Cloud, 2024) and LLMComparator (Kahng et al., 2024) explores the current state of human powered LLM qualitative evaluation through interviews with data analysts. These works find that practitioners often eyeball individual examples to interpret and look at qualitative differences between the outputs of two models, and develop an interactive web based application for users to inspect side-by-side LLM outputs with an LLM based rationale as to why one output is preferred over another. While these works are focused more on software tools rather than a pipeline which can be quantitavely verified, these HCI findings inform VibeCheck's vibe discovery mechanism to align with the human-powered qualitative process. Moreover, many NLP works (Zheng et al., 2023; Verga et al., 2024; Li et al., 2023; Park

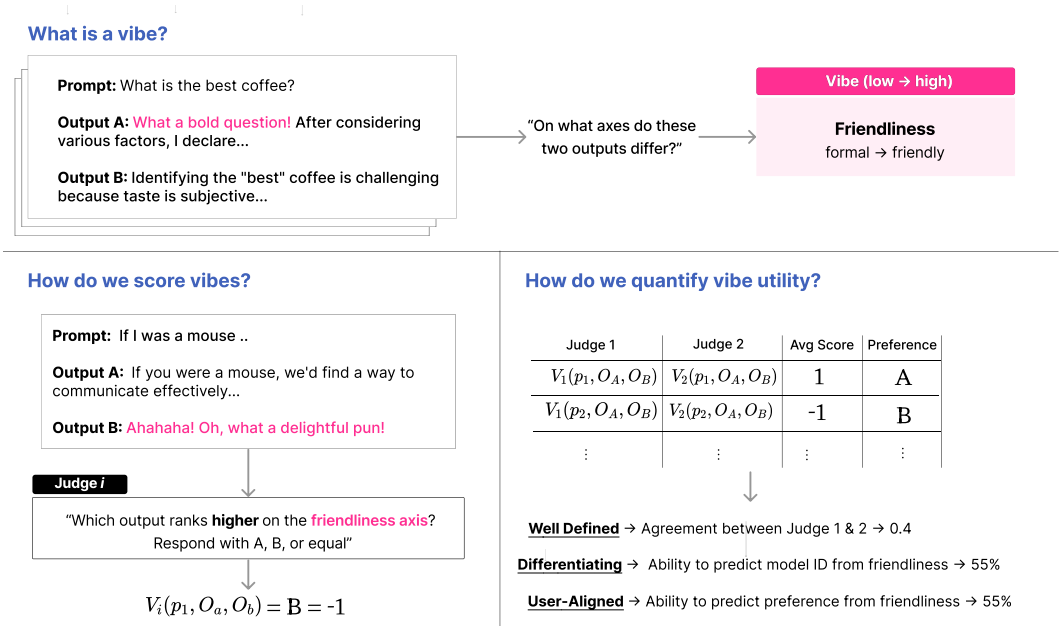

Figure 1: **Core components of VibeCheck.** A vibe is an axis along which a pair of outputs differ: for example, in the top panel, output A is more friendly while output B is more formal, defining a friendliness vibe. To score a prompt output triplet, a panel of LLM judges are used to determine which output falls higher on the vibe, resulting in a score of 1 (A), -1(B), or 0(tie). Finally, the scores obtained over a large set of outputs along with preference labels are used to compute vibe utility.

et al., 2024; Liusie et al., 2024) have explored using LLMs to predict user preference given responses from two models, showing these preference predictions often align with the judgements of human annotators. While these efforts focus more on the user experience, it does not provide an interpretable view of exactly *why* these users prefer one output over the other.

Discovering separable traits in unstructured data. In parallel to works in the machine learning community on LLM evaluation, there has been fantastic efforts in the HCI community on comparing generative model outputs as well as on using LLMs for qualitative analysis. Works like Torii et al. (2024); Byun et al. (2023) use LLMs to generate discussions from qualitative research data to automate the data analysis process, but note the lack of comprehensive evaluation metrics. Automated data analysis on unstructured data has also been explored in Zhong et al. (2022; 2023); Dunlap et al. (2024b), which use LLMs and VLMs to propose and validate candidate differences between two sets of text or images in the form of "set A contains more X", and Chiquier et al. (2024) employs an evolutionary algorithm to find text descriptions which best separates image classes to assist in zero-shot classification. We extend these works to pairwise inputs and introduce metrics of success which can better verify the separability, consistency, and alignment of these differences.

3 VIBE-BASED EVALUATIONS

We define a *vibe* as an axis along which a pair of texts can differ (e.g., "formal → friendly") that is perceptible to humans. A vibe ν is represented by a text description of the axis along with a definition of what it means to be high or low on this axis (e.g. "Tone: low = formal, high = friendly", see Figure 1). Identifying vibes aids users in selecting models that best suit their specific tasks. In this work, we focus on comparing the vibes of two models by discovering the axes on which their outputs differ and quantifying the utility of these vibes.

Consider a dataset D composed of triples $(\mathrm{p}, \mathrm{o}_A^{\mathrm{p}}, \mathrm{o}_B^{\mathrm{p}})$ and preference labels y_{p}, where p is a prompt and $\mathrm{o}_i^{\mathrm{p}}$ are the outputs from models A and B. For each triple, a judge (human or LLM) assigns a score for vibe ν, denoted $\nu(\mathrm{p}, \mathrm{o}_A^{\mathrm{p}}, \mathrm{o}_B^{\mathrm{p}}) \in \{-1, 0, 1\}$, which indicates whether model A scores lower (-1), similarly (0), or higher (1) than model B on this vibe. Thus, a vibe imposes an ordering on model outputs.